

Department Overview • Department Name: Department of Petroleum and Gas Engineering • Faculty: Faculty of Engineering & Architecture • History: Established in the Fall of 2004 as the pioneering department of the newly formed engineering faculty to address the industrial need for managing Balochistan's extensive natural oil and gas reserves (such as the Sui gas fields). • Accreditation: The four-year undergraduate program is fully accredited by the Pakistan Engineering Council (PEC) and follows the Outcome-Based Assessment (OBA) guidelines. • Core Focus: Equips students with scientific, mathematical, and computer-based skills for the exploration, drilling, appraisal, and optimal development of oil and gas assets.

Mission The program cultivates industrious engineers for the national and international petroleum and gas industry and its related fields in education, business, and government by providing technical knowledge, skills, and values through academic inquiry, teaching, and technical services.

Program Educational Objectives (PEOs) Graduates of the program are expected to achieve the following milestones within 3-5 years of graduation:

1. PEO-1: Address complex problems in fields like renewable and non-renewable energy and sustainable development by utilizing professional, teamwork, communication, and managerial skills.
2. PEO-2: Fulfill the requirements and necessary skills for higher academic studies and cutting-edge research.
3. PEO-3: Adopt a healthy and symbiotic environment with a strong commercial instinct, helping create new opportunities, especially in Balochistan.

Bachelor of Science in Petroleum & Gas Engineering (BS PGE)

Admission Requirements • Academic Background: F.Sc (Pre-Engineering) from a recognized board or equivalent with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in a relevant field with at least 60% marks. • Testing: An entry test administered by NTS.

Degree Requirements • Total Credit Hours: 136 • Total Courses: 44 • Minimum Academic Standing: CGPA $\geq$ 2.0 • PLO Requirement: Students must successfully pass all 12 standard Program Learning Outcomes (ranging from Engineering Knowledge and Problem Analysis to Ethics and Lifelong Learning).

Curriculum Plan (Program Schema by Semester) Note: Credit hour structures denote (Theory + Lab = Total).

1st Semester • PGE-102 | Fundamentals of Petroleum Engineering | Prerequisite: None | Credit Hours: 2+0=2 • GEO-201 | Applied Geology | Prerequisite: None | Credit Hours: 2+1=3 • HUM-101/112 | Islamic Studies/Ethics | Prerequisite: None | Credit Hours: 2+0=2 • MATHA 111 | Applied Mathematics-I | Prerequisite: None | Credit Hours: 3+0=3 • PHY-205 / HUM-163 | Applied Physics / Functional English (English-I) | Prerequisite: None / None | Credit Hours: 3+1=4 / 3+0=3 • ENGG 202L | Workshop Practice | Prerequisite: None | Credit Hours: 0+2=2

2nd Semester • HUM-268 / HUM-102 | Communication Skills (Eng-II) / Pakistan Studies | Prerequisite: None / None | Credit Hours: 3+0=3 / 2+0=2

3rd Semester • MATHA-116 | Applied Mathematics-II | Prerequisite: Applied Mathematics-I | Credit Hours: 3+0=3 • CHE-102 | Applied Chemistry | Prerequisite: None | Credit Hours: 2+1=3 • ENGG-201L | Engineering Drawing | Prerequisite: None | Credit Hours: 0+2=2 • CS-113 | Introduction to Computer Programming | Prerequisite: None | Credit Hours: 2+1=3 • HUM-276 |

Technical Report Writing & Pres. Skills (Eng-III) | Prerequisite: None | Credit Hours: 2+0=2 •
MATHA-211 | Applied Mathematics-III | Prerequisite: Applied Mathematics-II | Credit Hours:
3+0=3 • ENGG-206/211 | Fluid Mechanics / Principles of Electrical Eng. | Prerequisite: None /
None | Credit Hours: 2+1=3 / 2+1=3 • PGE-306 / ENGG-215 | Petroleum Geology & Geophysical
Exploration / Thermo | Prerequisite: None / None | Credit Hours: 3+0=3 / 2+1=3
4th Semester • STAT-203 / GEOE-206 | Applied Statistics / Mechanics of Materials | Prerequisite:
None / None | Credit Hours: 3+0=3 / 2+1=3 • PGE-303 / PGE-302 | Drilling Engineering-I /
Petrophysics | Prerequisite: None / None | Credit Hours: 3+1=4 / 3+1=4 • HUM-206 | Introduction
to Sociology | Prerequisite: None | Credit Hours: 3+0=3
5th Semester • MATHA-212 | Applied Numerical Methods | Prerequisite: None | Credit Hours:
2+1=3 • PGE-305 / PGE-402 | Properties of Reservoir Fluids / Drilling Engineering-II |
Prerequisite: None / Drilling Eng-I | Credit Hours: 3+1=4 / 3+1=4 • ENGG-322 / CHEE-313 |
Environment & Safety Management / Process Control | Prerequisite: None / None | Credit Hours:
3+0=3 / 2+1=3
6th Semester • PGE-311 / PGE-304 | Reservoir Geomechanics / Well Logging | Prerequisite:
None / Petrophysics | Credit Hours: 2+0=2 / 2+1=3 • PGE-309 / PGE-308 | Reservoir
Engineering-I / Petroleum Production Eng-I | Prerequisite: None / None | Credit Hours: 3+1=4 /
3+1=4 • CHEE-416 | Petroleum Refinery Engineering | Prerequisite: None | Credit Hours: 3+1=4
7th Semester • PGE-408L | Final Year Project (Part I) | Prerequisite: None | Credit Hours: 0+3=3 •
MGMT-402 | Project Planning and Management | Prerequisite: None | Credit Hours: 2+0=2 •
PGE-404 / PGE-405 | Well Testing / Petroleum Production Engineering-II | Prerequisite: None /
Petroleum Prod-I | Credit Hours: 3+1=4 / 3+1=4 • PGE-312 | Reservoir Simulation | Prerequisite:
None | Credit Hours: 3+1=4
8th Semester • PGE-408L | Final Year Project (Part II) | Prerequisite: None | Credit Hours: 0+3=3 •
PGE-412 / ECON-105 | Natural Gas Engineering / Economics | Prerequisite: None / None | Credit
Hours: 2+1=3 / 2+0=2 • PGE-407 / PGE-403 | Principles of Enhanced Oil Recovery / Reservoir
Engineering | Prerequisite: None / None | Credit Hours: 3+1=4 / 3+1=4